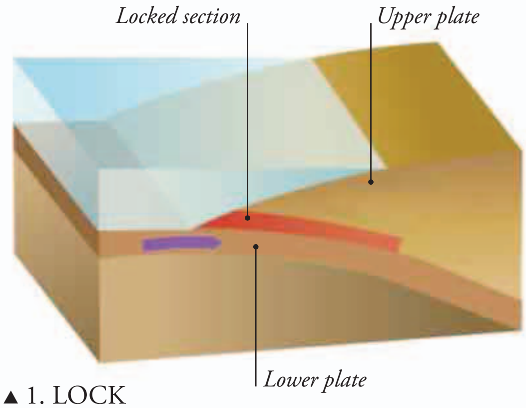
Tsunamis

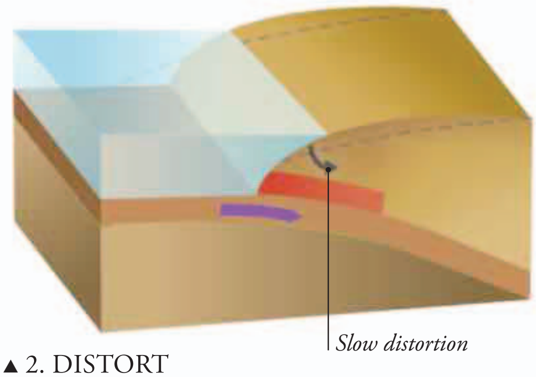
Every few years, a big earthquake on the ocean floor causes a massive rock movement that is transferred to the water and generates giant waves, known as tsunamis. Out on the open ocean, the waves are broad and low, covering a huge area. But when a tsunami reaches shallower water, the wave piles up like an extra-high tide that surges ashore and floods the land in just a few minutes. These waves are incredibly destructive and deadly.

WHY TSUNAMIS HAPPEN

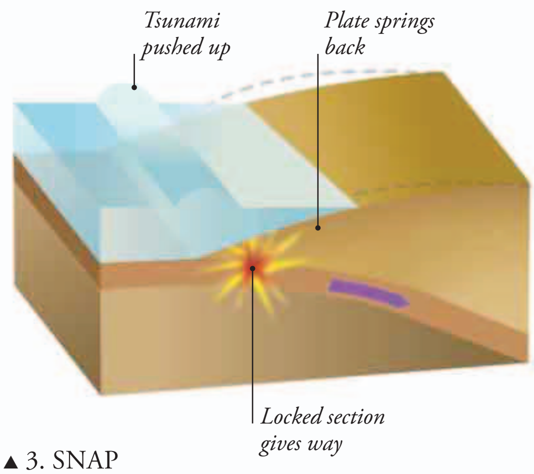
The biggest recent tsunamis have occurred in places where one plate of ocean floor is slipping beneath another, as shown below. The plates became locked together, then suddenly gave way, triggering oceanic earthquakes. But tsunamis can also be caused by volcanic eruptions, coastal landslides, and even collapsing ice shelves.



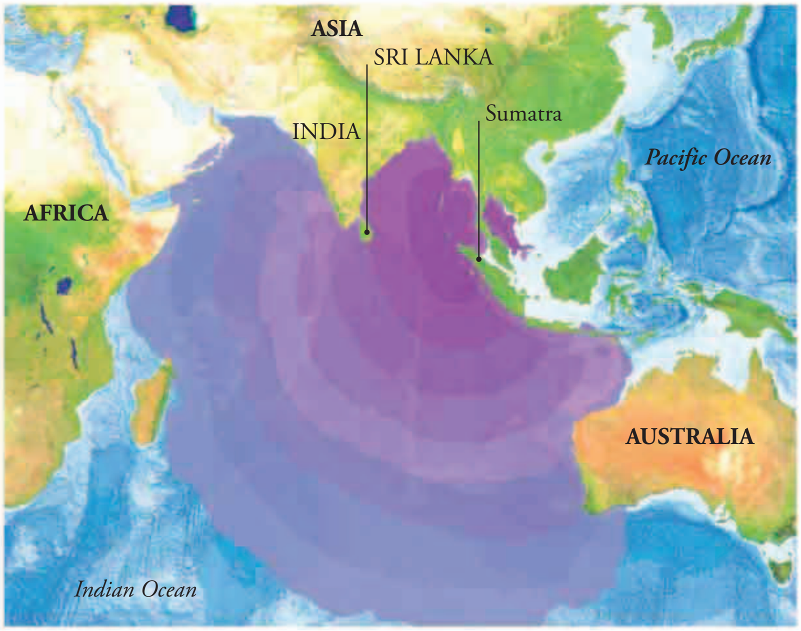
▲ 1. LOCK
As the lower plate pushes beneath the upper one, the sliding rocks become locked at the plate boundary.



▲ 2. DISTORT
Eventually, the moving lower plate bends the edge of the upper plate downward.



▲ 3. SNAP
When the tension makes the rocks give way, the edge of the top plate springs upward, pushing the water up into a giant heap that becomes a tsunami.



▲ 2004 TSUNAMI WAVES
Each color band on this map of the Indian Ocean shows the distance the tsunami waves traveled in one hour. The waves even reached the coast of Antarctica, although they were only about 3 ft (1 m) high by the time they got there.

RACING WAVES

Tsunami waves race across the ocean at incredibly high speeds. In December 2004, an earthquake off the northern tip of Sumatra caused a catastrophic tsunami that traveled outward across the Indian Ocean. It struck India and Sri Lanka just two hours later. This means that it traveled at about 500 mph (800 kph).